(Some) Vector & tensor algebra operations

1. Transpose

$$\text{If} \quad \underline{\underline{T}} = T_{ij} \quad \text{then} \quad T_{ji} = \underline{\underline{T}}^t \quad \left[\text{understood as } (\underline{\underline{T}}^t)_{ij \text{ entry}}\right]$$

Remember: flipping the indices creates a *new* tensor ("the transpose"); $T_{ij} = T_{ji}$ *only* for *symmetric* tensors! (**Note:** Panton denotes transpose with ^{t}, but you might also see $^\top$.)

2. Multiplication

2.1. vector/vector (dot product; row vector × column vector = scalar) – *only one that commutes*

$$\underline{u} \cdot \underline{v} = u_i v_i = v_i u_i = \underline{v} \cdot \underline{u}$$

Note: For $\underline{v} \cdot \underline{v} = v_i v_i$ **don't** write v_i^2 because this expression has a free index, and it's therefore a vector!

2.2. vector/tensor (row vector × matrix = row vector, or matrix × column vector = column vector) – *does not commute*

(**Note:** Panton puts a · between vectors and tensors for clarity; optional when using 1 vs. 2 underlines.)

$$\underline{v}\,\underline{\underline{T}} = v_i T_{ij} \qquad \text{or} \qquad (\underline{v}\,\underline{\underline{T}})_{j\text{th entry}} = \sum_{i=1}^{3} T_{ij} v_i$$

but we're allowed to (carefully!) re-index as

$$(\underline{v}\,\underline{\underline{T}})_{i\text{th entry}} = \sum_{j=1}^{3} T_{ji} v_j$$

which **does not commute** because

$$\underline{\underline{T}}\,\underline{v} = v_j T_{ij} \qquad \text{or} \qquad (\underline{\underline{T}}\,\underline{v})_{i\text{th entry}} = \sum_{j=1}^{3} T_{ij} v_j$$

Note: If we always write T_{ij}, then you can remember what index to put on v by whether $\underline{v}$ is in-front of $\underline{\underline{T}}$ (then i, and $\underline{v}$ is a row vector) or behind $\underline{\underline{T}}$ (then j, and $\underline{v}$ is a column vector) in the symbolic notation expression.

The following useful fact follows: $\underline{v}\,\underline{\underline{T}}$ is the row vector with the same entries as the column vector $\underline{\underline{T}}^t\,\underline{v}$.

2.3. vector/vector (dyadic product; column vector × row vector = matrix) – *does not commute*

$$\underline{u}\,\underline{v} = u_i v_j$$

does not commute because

$$\underline{v}\,\underline{u} = v_i u_j = u_j v_i$$

But, if we let $\underline{\underline{T}} = \underline{u}\,\underline{v}$, then we have shown that $\underline{v}\,\underline{u} = \underline{\underline{T}}^t$.
Sometimes we denote "$\underline{u}\,\underline{v}$" as "$\underline{u} \otimes \underline{v}$" (so we don't accidentaly confuse it for "$\underline{u} \cdot \underline{v}$").

2.4. tensor/tensor (direct product; matrix × matrix = matrix) – *does not commute*

$$\underline{\underline{R}} = \underline{\underline{T}}\,\underline{\underline{S}} \quad \text{or} \quad R_{ij} = T_{ik} S_{kj} \quad \Rightarrow \quad \text{(for example)} \quad T_{ik} S_{jk} = \underline{\underline{T}}\,\underline{\underline{S}}^t \neq \underline{\underline{R}}\,!$$

This is simply multiplication of matrices, where k runs over the ith row of $\underline{\underline{T}}$ *and* jth column of $\underline{\underline{S}}$:

$$R_{ij} = \sum_{k=1}^{3} T_{ik} S_{kj}$$

"Row times column" ⇒ inner indices (2nd one on T and 1st one on S, in index notation) **must match!**

2.5. tensor/tensor (double contraction; "dot product" for tensors) – *does not commute*

$$\underline{\underline{T}} : \underline{\underline{S}} = T_{ij} S_{ji} \quad \Rightarrow \quad \text{(for example)} \quad T_{ij} S_{ij} = \underline{\underline{T}} : \underline{\underline{S}}^t \neq \underline{\underline{T}} : \underline{\underline{S}}$$

Note: If we define $R_{kl} = T_{kj} S_{jl}$ then $\mathrm{tr}(\underline{\underline{R}}) = R_{kk} = R_{ii} = T_{ij} S_{ji}$, in other words $\underline{\underline{T}} : \underline{\underline{S}} = \mathrm{tr}(\underline{\underline{T}}\,\underline{\underline{S}})$.